

DHCP and Relay

```
R1
[OK]
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip dhcp pool hr
R1(dhcp-config)#network 50.0.0.0 255.0.0.0
R1(dhcp-config)#default-router 50.0.0.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 1 2 3
R1(dhcp-config)#exit
R1(config)#ip dhcp excluded-address 50.0.0.1 50.0.0.50
R1(config)#exit
R1#
Sep  2 07:13:09.604: %SYS-5-CONFIG_I: Configured from console by console
R1#wr
Building configuration...
[OK]
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#do wr
Building configuration...
[OK]
R1(config)#exit
R1#
Sep  2 07:13:26.744: %SYS-5-CONFIG_I: Configured from console by console
```

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R1
Sep  2 07:18:51.221: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type
                Hardware address/
                User name
50.0.0.51       0100.5079.6668.37  Sep 03 2026 02:49 PM  Automatic
R1#
R1#
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip dhcp pool sales
R1(dhcp-config)#network 192.168.2.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.2.1
R1(dhcp-config)#dns-server 50.0.0.30
R1(dhcp-config)#do wr
Building configuration...
[OK]
R1(dhcp-config)#exit
R1(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.50
R1(config)#do wr
Building configuration...
[OK]
R1(config)#exit
```

```
R2
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

R2#
R2#show ip route
50.0.0.0/8 [1/0] via 172.16.0.1
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, Ethernet0/1
L       172.16.0.2/32 is directly connected, Ethernet0/1
C       192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, Ethernet0/0
L       192.168.2.1/32 is directly connected, Ethernet0/0
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)# int e0/0
R2(config-if)#ip helper-address 172.16.0.1
R2(config-if)#
```

```
VPC-1
Modified version for EVE-NG.

Press '?' to get help.

Executing the startup file

VPC-1> ip dhcp
DDORA IP 50.0.0.51/8 GW 50.0.0.1

VPC-1> show ip

NAME          : VPC-1[1]
IP/MASK       : 50.0.0.51/8
GATEWAY       : 50.0.0.1
DNS           : 50.0.0.30
DHCP SERVER   : 50.0.0.1
DHCP LEASE    : 91945, 93780/46890/82054
MAC           : 00:50:79:66:68:37
LPORT        : 20000
RHOST:PORT    : 127.0.0.1:30000
MTU           : 1500

VPC-1>
```

```
VPC-2
GATEWAY       : 0.0.0.0
DNS           :
MAC           : 00:50:79:66:68:a5
LPORT        : 20000
RHOST:PORT    : 127.0.0.1:30000
MTU           : 1500

VPC-2> ip dhcp
DDORA IP 50.0.0.52/8 GW 50.0.0.1

VPC-2> show ip

NAME          : VPC-2[1]
IP/MASK       : 50.0.0.52/8
GATEWAY       : 50.0.0.1
DNS           : 50.0.0.30
DHCP SERVER   : 50.0.0.1
DHCP LEASE    : 93777, 93780/46890/82054
MAC           : 00:50:79:66:68:a5
LPORT        : 20000
RHOST:PORT    : 127.0.0.1:30000
MTU           : 1500

VPC-2>
```

```
VPC-4
Modified version for EVE-NG.

Press '?' to get help.

Executing the startup file

VPC-4> ip dhcp
DDORA IP 192.168.2.52/24 GW 192.168.2.1

VPC-4> show ip

NAME          : VPC-4[1]
IP/MASK       : 192.168.2.52/24
GATEWAY       : 192.168.2.1
DNS           : 50.0.0.30
DHCP SERVER   : 172.16.0.1
DHCP LEASE    : 85429, 86400/43200/75600
MAC           : 00:50:79:66:68:a6
LPORT        : 20000
RHOST:PORT    : 127.0.0.1:30000
MTU           : 1500

VPC-4>
```

```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    Autoconfiguration IP Address. . . : 169.254.97.86
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 

C:\Documents and Settings\User>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 192.168.2.51
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.2.1

C:\Documents and Settings\User>
```